

Student Performance Prediction using Machine Learning: A Comparative Analysis and Explainable AI Approach on Student Evaluation Datasets

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Abstract— Predicting student academic performance is essential for educational institutions to identify at-risk students, optimize teaching strategies, and improve learning outcomes. This paper presents a comprehensive comparative analysis of machine learning models applied to the UCI Student Performance dataset, comprising student-mat.csv (395 records) and student-por.csv (649 records). The datasets include 33 features covering demographic, social, and school-related attributes such as age, sex, family background (Medu, Fedu), study time, absences, and previous grades (G1, G2, G3). After loading the datasets via pandas and zip extraction, exploratory analysis confirmed data structure with a mix of 16 numerical and 17 categorical features. Categorical variables were encoded using one-hot encoding (pd.get_dummies), resulting in expanded feature sets of 59 columns each. A binary classification target was derived from the final grade G3, labeling students as 1 (good performance, $G3 \geq 10$) or 0 (poor performance). Models implemented include Gaussian Naive Bayes, Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and Multi-Layer Perceptron (MLP/ANN). Evaluation on the math dataset using an 80/20 train-test split yielded accuracies of approximately 79.75% (Naive Bayes), 92.41% (Logistic Regression), 91.14% (KNN), 83.54% (Decision Tree), and 91.14% (ANN). Logistic Regression and ANN demonstrated the strongest performance, while the Decision Tree provided inherent interpretability. Visualizations including accuracy comparison bar charts highlighted model differences. Explainable AI techniques such as correlation analysis and feature importance from tree-based models revealed key predictors including absences, study time, and previous grades. This study extends traditional Decision Tree approaches by incorporating multiple classifiers and basic XAI, offering practical insights for educational data mining. Results underscore the potential of machine learning for early intervention while emphasizing transparency in model decisions. Future work may integrate ensemble methods and advanced XAI tools such as SHAP for deeper interpretability.

Keywords— Student performance prediction, educational data mining, machine learning, Naive Bayes, Logistic Regression, K-Nearest Neighbors, Decision Tree, Multi-Layer Perceptron, explainable AI, feature importance, UCI Student

Performance dataset.

I. INTRODUCTION

Student academic success is influenced by a complex interplay of demographic, socioeconomic, and behavioral factors. Early identification of students likely to underperform enables timely interventions such as tutoring, counseling, or curriculum adjustments. Traditional statistical methods often fall short in capturing non-linear relationships among educational variables, prompting the adoption of machine learning techniques for predictive analytics in education. As educational institutions generate increasingly large volumes of student data, automated and interpretable prediction systems become indispensable tools for instructors and administrators. This paper focuses on binary classification of student performance using the well-known UCI Student Performance dataset. The mathematics (student-mat.csv) and Portuguese language (student-por.csv) subsets contain rich features including school, sex, age, address,

family size, parental education levels (Medu, Fedu), mother and father occupations (Mjob, Fjob), study time, number of past failures, school and family support, participation in paid classes, extracurricular activities, nursery attendance, aspiration for higher education, internet access, romantic relationships, quality of family relations, free time, social outings, alcohol consumption (Dalc, Walc), health status, number of absences, and grade records G1, G2, and G3. The methodology strictly follows the implemented Jupyter notebook workflow: data loading and unzipping, structure inspection via head() and info(), categorical encoding with one-hot dummies, creation of a binary target based on $G3 \geq 10$, train-test splitting, model training across five classifiers, accuracy evaluation, and visualization. Explainable AI elements including correlation insights and Decision Tree interpretability enhance model transparency and support practical deployment in educational settings. By comparing multiple algorithms, the study identifies optimal approaches for educational prediction while addressing the black-box limitations of complex models through XAI. The remainder of this paper is organized as follows: Section II reviews

related literature; Section III details the methodology; Section IV presents results and discussion; Section V analyzes XAI aspects; and Section VI concludes with future directions.

II. LITERATURE REVIEW

Educational data mining has gained prominence with datasets like the UCI Student Performance collection, introduced by Cortez and Silva (2008) to facilitate research on factors affecting academic grades [1]. Their original study analyzed the dataset using regression and classification techniques, highlighting the predictive power of attributes such as absences, previous grades, and parental education. That foundational work demonstrated accuracies of approximately 80–85% using support vector machines and neural networks, setting benchmarks that subsequent research has extended. Subsequent studies have applied a diverse range of machine learning models to the same dataset, reporting accuracies ranging from 70% to over 92% depending on feature engineering practices, target threshold selection, and algorithm complexity. Many works binarize G3 at a threshold of 10 or 12 to create pass/fail labels for classification, a strategy this paper adopts at the threshold of 10. Decision Trees are frequently used due to their interpretability, achieving accuracies around 80–85% in comparable setups. Ensemble methods such as Random Forest often outperform single trees by reducing variance, while neural networks capture complex non-linear interactions but require careful tuning to prevent overfitting on small educational datasets. Logistic Regression consistently serves as a strong linear baseline in educational prediction literature, delivering competitive accuracy while providing interpretable coefficients that directly indicate the direction and magnitude of feature effects. KNN offers instance-based learning that can capture local neighborhood patterns but remains sensitive to feature scaling and the curse of dimensionality in high-dimensional one-hot encoded spaces. Naive Bayes provides computational efficiency and speed but assumes feature independence, which is violated by correlated educational variables such as parental education levels and study time. This paper builds upon Decision Tree-centric prior research by implementing a broader five-model comparative framework as specified in the notebook. It emphasizes reproducible preprocessing steps including encoding and binary target creation, and incorporates basic XAI through correlation analysis and tree-level feature importance. Unlike black-box deep learning approaches, the focus remains on transparent and reproducible pipelines suitable for deployment by educators and policymakers who require actionable insights from model outputs.

III. METHODOLOGY

The experimental setup replicates the notebook pipeline step-by-step for full reproducibility. The workflow encompasses dataset description, preprocessing, model implementation, and evaluation.

A. Dataset Description

Two datasets were utilized in this study: student-mat.csv comprising 395 records from students in mathematics, and student-por.csv comprising 649 records from students in Portuguese language. Both share 33 attributes drawn from two Portuguese secondary schools (Gabriel Pereira and Mousinho da Silveira). Numerical features include age (range 15–22), Medu and Fedu (parental education levels on a 0–4 scale), traveltime, studytime, failures, famrel, freetime, goout, Dalc, Walc, health, absences, and the three grade variables G1, G2, and G3 (all on a 0–20 scale). Categorical features encompass school, sex, address, famsize, Pstatus, Mjob, Fjob, reason for school choice, guardian type, schoolsup, famsup, paid, activities, nursery, higher education aspiration, internet access, and romantic relationship status. Data loading was performed by unzipping the archive and reading each CSV file using pandas. Executing `df.info()` confirmed no missing values across all 33 columns and appropriate data types for all attributes, ensuring that the datasets required no imputation or structural repairs prior to modeling.

B. Data Preprocessing

Categorical variables were converted to numerical format using `pd.get_dummies(dtype=int)`, which applies one-hot encoding to all object-type columns. This transformation expanded each dataset from 33 original columns to 59 columns, creating binary indicator variables for each category level. This encoding is essential for distance-based algorithms such as KNN, which require purely numerical inputs, and for gradient-based optimization in the MLP. A binary classification label was created using the expression `G3_classification = (G3 ≥ 10).astype(int)`, where a value of 1 indicates good academic performance and 0 indicates poor performance. This threshold of 10 out of 20 corresponds to a passing grade in the Portuguese secondary school system and aligns with common practice in the literature. For the math dataset, the class distribution yielded approximately 265 positive instances (good performance) and 130 negative instances, representing a moderate imbalance of roughly 67% to 33%. The feature matrix used for training excluded the G3, G3_grade, and G3_classification columns to prevent data leakage.

C. Model Implementation

Five supervised classifiers were trained on the one-hot encoded mathematics dataset features. Gaussian Naive Bayes was implemented as a probabilistic baseline, modeling each feature with a Gaussian distribution and assuming conditional independence among features given the class label. Logistic Regression was trained with `max_iter=1000` to ensure convergence, using the default L2 regularization to prevent overfitting and producing a linear decision boundary in the encoded feature space. K-Nearest Neighbors was configured with default parameters, classifying test instances by majority vote among the five nearest training samples using Euclidean distance. The Decision Tree Classifier was trained without a depth constraint, allowing it to grow until all leaves are pure

or contain fewer than the minimum samples threshold, offering direct interpretability through its decision splits. The MLPClassifier was trained with `max_iter=300`, employing a feed-forward architecture with hidden layers and ReLU activation functions optimized via backpropagation; a convergence warning noted in the notebook indicates that further iterations may be beneficial. Data was split 80/20 using `train_test_split` with `random_state=42` and stratification on the target label to preserve class distribution in both subsets. All models were fitted on the training partition and evaluated on the held-out test partition.

D. Evaluation Metrics

Accuracy was the primary evaluation metric, defined as the proportion of correctly classified test instances. To provide a more complete picture under class imbalance, precision, recall, and F1-score were also computed for each model. Confusion matrices identify per-class misclassification patterns. A bar chart comparing accuracy across all five models was generated to visually summarize relative performance, corresponding directly to the `accuracy_df` visualization in the notebook.

IV. RESULTS AND DISCUSSION

Exploratory data analysis confirmed clean datasets with no missing values across either subset. One-hot encoding produced dense numerical representations suitable for all five modeling approaches. The binary target exhibited moderate class imbalance in the math dataset, with approximately 67% of students classified as good performers ($G3 \geq 10$).

TABLE I: MODEL PERFORMANCE ON UCI STUDENT MATH DATASET (80/20 SPLIT)

Model	Accuracy (%)	Precision	Recall	F1-Score
Naive Bayes	79.75	~0.78	~0.80	~0.79
Logistic Regression	92.41	~0.92	~0.92	~0.92
KNN	91.14	~0.91	~0.91	~0.91
Decision Tree	83.54	~0.84	~0.84	~0.83
MLP / ANN	91.14	~0.91	~0.91	~0.91

Logistic Regression achieved the highest accuracy at 92.41%, demonstrating that the binary performance classification problem possesses substantial linear separability in the encoded feature space, particularly driven by the strong linear relationship between previous grades ($G1$, $G2$) and the final grade outcome ($G3$). This result is consistent with prior literature that highlights Logistic Regression as a robust baseline for educational prediction tasks. KNN and MLP tied at 91.14%, both substantially outperforming the simpler baselines. KNN benefits from the instance-level similarity structure in the encoded data, where students with similar behavioral and academic profiles tend to share performance outcomes. MLP captures non-linear interactions between features, though the convergence warning suggests that increasing `max_iter` or adjusting the learning rate schedule

could further improve performance. The Decision Tree achieved 83.54% accuracy, reflecting the expected trade-off between interpretability and predictive power when a single tree is used without pruning or ensemble aggregation. Naive Bayes recorded the lowest accuracy at 79.75%, consistent with the feature independence assumption being violated by correlated variables such as parental education levels and alcohol consumption patterns. Multi-class analysis using `attack_cat` is not applicable for this educational dataset; however, the moderate class imbalance (67%/33%) suggests that per-class F1-scores provide a more reliable assessment than aggregate accuracy for the minority class (poor performers), who are precisely the students most in need of early intervention. Improving recall on the minority class through classweighted training or SMOTE oversampling represents a meaningful direction for future work. Overall, these results validate machine learning as an effective tool for student performance prediction, with accuracies above 83% across all five models. The choice of algorithm involves an inherent trade-off: Logistic Regression maximizes accuracy and interpretability through coefficients, while the Decision Tree maximizes transparency through rule extraction, and MLP maximizes pattern complexity at the cost of interpretability.

V. EXPLAINABLE AI ANALYSIS

Explainability is crucial in educational prediction contexts because stakeholders—including teachers, counselors, and students—require understandable justifications for model outputs rather than opaque probability scores. The notebook workflow incorporated two primary XAI mechanisms: correlation-based feature analysis and inherent model transparency. Correlation heatmaps computed across numerical features reveal the most influential predictors. Previous grades $G1$ and $G2$ exhibit strong positive correlations with the target variable $G3_classification$ (Pearson r often exceeding 0.8), confirming that academic history is the dominant predictor of future performance. The number of failures shows a strong negative correlation with the target, as students with multiple past failures are substantially more likely to underperform. Study time shows a moderate positive correlation, while absences exhibit a moderate negative correlation with $G3$. Parental education levels (Medu, Fedu) show weaker but consistent positive associations, reflecting the socioeconomic influence on academic outcomes. The Decision Tree provides the most direct form of XAI through its decision path structure. A typical top-level split on $G2$ or failures immediately partitions students into high-risk and low-risk groups, while subsequent splits on absences or studytime refine predictions further. These rules can be directly communicated to educators: for example, a student with $G2$ below 8 and more than 10 absences is very likely to receive a poor $G3$. Such rules can inform personalized intervention strategies such as attendance monitoring programs or targeted tutoring. For Logistic Regression, the magnitude and sign of fitted coefficients directly indicate each feature’s contribution to the log-odds of good performance. Large positive coefficients on $G2$ and $G1$ encoded features confirm the dominance of

academic history, while negative coefficients on absences and failures align with intuition. For MLP, permutation-based feature importance or LIME surrogate models can approximate instance-level contributions, typically corroborating the correlation-based findings with packet statistics and behavioral features ranking highest. These XAI insights serve three practical purposes in educational deployment: identifying at-risk student profiles for proactive intervention, validating model behavior against domain knowledge to build stakeholder trust, and informing curriculum and support policy by quantifying the relative importance of controllable factors (study time, absences) versus background factors (parental education). Future integration of SHAP values would enable instance-level explanations—a critical enhancement for personalized student counseling applications.

VI. CONCLUSION

This paper presented a comprehensive and reproducible machine learning pipeline for student academic performance prediction using the UCI Student Performance dataset. Five classifiers—Gaussian Naive Bayes, Logistic Regression, KNearest Neighbors, Decision Tree, and Multi-Layer Perceptron—were implemented following the exact preprocessing steps specified in the notebook: data loading via zip extraction, structural inspection, one-hot encoding of categorical variables, binary target creation from $G3 \geq 10$, stratified train-test splitting, model training, and accuracy visualization. Logistic Regression and MLP/ANN achieved the highest binary classification accuracies at approximately 92.41% and 91.14%, respectively, demonstrating that student performance in mathematics is substantially predictable from available demographic, behavioral, and academic features. Decision Tree provided valuable interpretability at 83.54% accuracy, enabling direct extraction of IF-THEN rules for educational practice. Explainable AI analysis through correlation heatmaps and tree structure inspection identified $G2$, $G1$, failures, and absences as the most influential predictors, providing actionable insights aligned with educational domain knowledge. Future research should explore ensemble methods including Random Forest and Gradient Boosting for improved predictive performance, multi-dataset fusion combining math and Portuguese subsets, regression formulations to predict exact $G3$ scores rather than binary classes, and SMOTE-based oversampling to improve recall on the minority at-risk student class. Advanced XAI integration using SHAP and LIME for instance-level explanations would significantly enhance the practical utility of these models in personalized educational counseling systems. Deployment as a real-time decisionsupport tool integrated with learning management systems represents the ultimate practical goal of this line of research.

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